

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

***CO2:** Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks:25

Annexure A

Analytical Problem Solving

Code of Conduct:

*I Deepan Kumar P (192312498) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is -0.5 . For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

Given:

- Study Hours (x_1) = 2
- Attendance (x_2) = 1
- Weight $w_1 = 0.6$
- Weight $w_2 = 0.4$
- Bias $b = -0.5$

Step 1: Calculate Net Input

$$\begin{aligned} Net &= (w_1x_1) + (w_2x_2) + b \\ &= (0.6 \times 2) + (0.4 \times 1) - 0.5 \\ &= 1.2 + 0.4 - 0.5 = 1.1 \end{aligned}$$

Step 2: Apply Step Activation Function

Since

$$Net = 1.1 > 0$$

Output = 1

Final Answer

- Net Input = 1.1
- Output = 1
- Conclusion: Student Passes

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

Given:

- Initial Weight = 0.5
- Learning Rate (η)=0.1
- Input (x)=2
- Actual Output =0.6
- Target Output =1

Step 1: Error

$$\begin{aligned} \text{Error} &= \text{Target} - \text{Actual} \\ &= 1 - 0.6 = 0.4 \end{aligned}$$

Step 2: Weight Update

$$\begin{aligned}\Delta w &= \eta \times Error \times Input \\ &= 0.1 \times 0.4 \times 2 \\ &= 0.08\end{aligned}$$

Step 3: Updated Weight

$$\begin{aligned}New\ Weight &= Old + \Delta w \\ &= 0.5 + 0.08 = 0.58\end{aligned}$$

Final Answer

- Error = **0.4**
- Weight Update = **0.08**
- New Weight = **0.58**

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors.

Solution

Given

- TP = 180
- TN = 150
- FP = 20
- FN = 50

$$Total = 180 + 150 + 20 + 50 = \mathbf{400}$$

Accuracy

$$= \frac{180 + 150}{400} = 82.5\%$$

Precision

$$= \frac{180}{180 + 20} = 90\%$$

Sensitivity (Recall)

$$= \frac{180}{180 + 50} = 78.26\%$$

Specificity

$$= \frac{150}{150 + 20} = 88.24\%$$

Answer

- Accuracy = **82.5%**
- Precision = **90%**
- Sensitivity = **78.26%**
- Specificity = **88.24%**

Conclusion:

The model has high precision and good accuracy. However, the recall is only **78.26%**, indicating that some disease cases are missed (false negatives). Since minimizing false negatives is critical in medical diagnosis, the model should be improved before deployment.

4.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Individual	Fitness
2	4
4	16
6	36
8	64

Best Candidates

- 8 (Fitness = 64)
- 6 (Fitness = 36)

Crossover

Binary:

6 = 0110

8 = 1000

Exchange the least significant bit (LSB):

- 0110 → 0111 = 7
- 1000 → 1001 = 8

New Offspring

- 7
- 8

Role of Mutation

Mutation randomly changes one or more bits in a chromosome. It increases population diversity, prevents premature convergence, helps escape local optima, and improves the chances of finding the global optimum.

5 .During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

Solution

Observation

- Training Accuracy: 60% → 95%
- Validation Accuracy: 58% → 72% → 68%

Issue Identified

The model is experiencing **Overfitting**.

Reason

The model learns the training data too well, including noise and irrelevant patterns. As a result, it performs poorly on unseen validation data.

Techniques to Reduce Overfitting

1. Early Stopping

2. Regularization (L2 Regularization or Dropout)

Importance of Validation Accuracy

- Measures model performance on unseen data.
- Detects overfitting.
- Helps in selecting the best model.
- Indicates the model's generalization ability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand